

What is Threat Intelligence

Threat Intelligence (TI) = Information about cyber threats (attackers, their methods, tools, infrastructure, and motives) that helps organizations **predict, detect, and respond** to attacks. It's like **knowing how thieves break into houses** → what tools they use, which doors they target, and when they usually attack → so you can strengthen your locks and monitor more effectively.

Why Do Companies Use Threat Intelligence?

Companies invest in TI because:

1 Early Warning System

- TI alerts about **emerging threats** before they hit your network.
- Example: If a phishing campaign is spreading in your industry, TI warns you to watch for certain domains/attachments.

2 Better Security Detection (SOC Benefit)

- TI provides **Indicators of Compromise (IOCs)** like malicious IPs, domains, hashes.
- SOC teams can **feed these into SIEM/EDR** tools to auto-detect attacks.

3 Understanding Attackers (Threat Actors)

- TI explains **who is attacking** (cyber criminals, nation-states, hacktivists) and **why** (money, espionage, disruption).
- Example: A bank might track ransomware groups targeting financial institutions.

4 Reduce Noise & Prioritize

- Not every alert is urgent.
- TI helps identify **real threats vs false positives**, saving analyst time.

5 Incident Response & Forensics

- During investigation, TI provides **context** (e.g., "This hash belongs to TrickBot malware family").
- Helps respond faster and smarter.

6 Compliance & Risk Management

- Many regulations (like ISO, NIST, GDPR) require **proactive threat monitoring**.
- TI shows auditors that the company is **serious about cyber defense**.

Types of Threat Intelligence

1 Strategic Threat Intelligence

- **Audience:** Executives, management, CISOs.
- **Purpose:** Big-picture view → *Who is attacking? Why are they targeting us? What's the risk to the business?*

- **Content:**
 - Threat actor groups (e.g., APT29, FIN7).
 - Their motives (financial, espionage, hacktivism).
 - Industry/region being targeted.
- **Example:**

“A ransomware group is increasingly targeting healthcare companies in Asia for financial gain.”
- **SOC benefit:** Justifies budgets, policy changes, and security investments.

2 Tactical Threat Intelligence

- **Audience:** SOC Analysts, Security Engineers.
- **Purpose:** Day-to-day defense → *How do attackers operate? What methods/tools do they use?*
- **Content:**
 - MITRE ATT&CK techniques (e.g., phishing, privilege escalation).
 - Tools & malware used (Mimikatz, Cobalt Strike, Emotet).
 - Campaign behavior patterns.
- **Example:**

“Phishing emails are using malicious macros in Word docs to drop Emotet malware.”
- **SOC benefit:** Helps analysts tune SIEM, IDS/IPS rules, and improve detection.

3 Operational Threat Intelligence

- **Audience:** Incident Response (IR) & Threat Hunting teams.
- **Purpose:** Real-time info → *What’s happening in this ongoing attack?*
- **Content:**
 - Specific TTPs observed during an attack.
 - Command-and-Control (C2) infrastructure details.
 - Attack timelines and attacker behavior.
- **Example:**

“Current attack is using PowerShell scripts to connect to C2 server at malicious[.]com.”
- **SOC benefit:** Helps respond faster by knowing attacker’s next move.

4 Technical Threat Intelligence

- **Audience:** Automated defense tools (SIEM, EDR, firewalls).
- **Purpose:** Raw machine-readable data for blocking/detection.
- **Content (IoCs):**
 - Malicious IPs
 - Domains/URLs
 - File hashes (MD5, SHA256)

- Email addresses used in phishing
- **Example:**
“Block IP 192.168.10.45, SHA256 hash of malware abc123..., and domain badmail[.]xyz.”
- **SOC benefit:** Feeds into SIEM/Firewall for automatic threat detection & blocking.

🔑 Quick Summary (Easy to Remember)

- **Strategic** → Who & Why (executive level).
- **Tactical** → How (methods & tools, for SOC).
- **Operational** → What’s happening now (IR & hunters).
- **Technical** → Exact IoCs (IPs, hashes, domains).

✅ Simple Example

Imagine your company gets lots of **phishing emails**.

- Without TI → You only react after employees report suspicious emails.
- With TI → You already know the **domains & IPs used in that campaign** → You block them in your firewall and email filter → Attack stopped before reaching inbox.

👉 In short:

Threat Intelligence = Turning raw cyber data into useful knowledge to stop attacks before they succeed.